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MENISCUS REPAIR IMPLANT AND METHOD OF USE

Abstract

A collagen scaffold may be used in repairing a meniscal tear within a patient's meniscus. A collagen scaffold may be secured relative to a delivery device. The collagen scaffold may be placed within the meniscal tear and may be sutured in position. The sutures may pass through the collagen scaffold. The collagen fibers of the collagen scaffold may be substantially aligned with collagen fibers of the meniscus. In some cases, the collagen scaffold may be delivered to the meniscal tear using a securement member that includes forks that are adapted to extend into the collagen scaffold prior to folding and then rolling up the collagen scaffold and securement member to create a scaffold assembly that can be disposed within a delivery device.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/US2023/080530, filed on Nov. 20, 2023, which claims the benefit of U.S. Patent Application Ser. No. 63/427,287 filed on Nov. 22, 2022, the disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure pertains generally, but not by way of limitation, to orthopedic implants and methods of treatment. More particularly, the present disclosure relates to a meniscus repair implant, such as one that is engineered for use within portions of the meniscus that have little to no native blood flow

BACKGROUND

[0003] The meniscus is part of the knee, and is a C-shaped piece of tough, rubbery cartilage that is made out of collagen and that acts as a shock absorber between the tibia and the femur. The meniscus includes a medial meniscus that is located on the inside of the knee and a lateral meniscus that sits on the outside of the knee. Meniscus tears are a common knee injury for athletes and non-athletes alike, and can occur when a patient twists or turns their upper leg while their foot is planted and their knee is bent. Treatment for meniscal tears can include simply cutting out the damaged portion of the meniscus. An advantage of this treatment is that the patient typically walks same day after surgery. A disadvantage of this treatment is that the patient ends up with less meniscal material, which does not replace itself. Another treatment for meniscus tears involves repairing the tear, which can be complicated by a relative lack of blood flow within the meniscus. An outer third of the meniscus is the “red” zone, which is vascularized and thus has the best healing potential. A middle third of the meniscus is the “red-white” zone, which has some blood flow. The inner third of the meniscus is the “white” zone, which lacks blood flow. Repairs to the “red-white” zone or the “red” zone may have poor healing results due to the lack of blood flow. Accordingly, there is an ongoing need to provide methods and materials for treating injuries to the meniscus.

BRIEF SUMMARY

[0004] This disclosure provides design, material, manufacturing method, and use alternatives for implants used for soft tissue repairs such as but not limited to meniscus repairs. An implant may include a collagen scaffold. An example may be found in a method for repairing a meniscal tear within a patient's meniscus. The method includes securing a meniscal implant relative to a delivery device to form a scaffold delivery assembly and delivering the scaffold delivery assembly to a position proximate the meniscal tear. The meniscal implant is inserted into the meniscal tear and is held within the meniscal tear by placing one or more fasteners extending across the meniscal tear.

[0005] Alternatively or additionally, the method may further include separating the meniscal implant from the delivery device after inserting the meniscal implant into the meniscal tear.

[0006] Alternatively or additionally, the method may further include intra-operatively cutting the meniscal implant to an appropriate size prior to securing the meniscal implant to the delivery device.

[0007] Alternatively or additionally, the one or more fasteners may include one or more sutures.

[0008] Alternatively or additionally, the one or more sutures may pass through the meniscal

implant.

[0009] Alternatively or additionally, the delivery device may include an elongate member, a cross member including a first cross member segment extending laterally from the elongate member in a first direction and a second cross member segment extending laterally from the elongate member in a second direction, and a plurality of fork members extending from the cross member, the plurality of fork members extending in a plane transverse to a plane extending through the elongate member and the cross member.

[0010] Alternatively or additionally, the method may further include intra-operatively imparting a bend to an elongate shaft of the delivery device prior to delivering the scaffold delivery assembly to the position proximate the meniscal tear.

[0011] Alternatively or additionally, the step of inserting the meniscal implant into the meniscal tear may include substantially aligning collagen fibers of the meniscal implant with collagen fibers of the meniscus.

[0012] Another example may be found in a method for delivering a treatment scaffold to a meniscal tear within a patient's meniscus. The method includes securing a treatment scaffold to a securement member adapted to releasably hold the treatment scaffold and loading the treatment scaffold and securement member into a delivery device. The treatment scaffold is deployed within the meniscal tear with the delivery device. The treatment scaffold is held within the meniscal tear by placing one or more sutures across the meniscal tear.

[0013] Alternatively or additionally, the delivery device may include an elongate shaft, and the method may further include bending the elongate shaft in order to align the treatment scaffold and securement member with the meniscal tear.

[0014] Alternatively or additionally, the delivery device may further include a handle, an actuatable trigger slidably disposed within the handle, and a sheath adapted to fit over the securement member, the sheath operably coupled with the actuator trigger such that moving the trigger proximally relative to the handle withdraws the sheath from the securement member.

[0015] Alternatively or additionally, the delivery device may further include a cannula extending over the delivery device, the cannula removable before deploying the treatment scaffold.

[0016] Alternatively or additionally, the method may further include intra-operatively cutting the treatment scaffold to an appropriate size prior to securing the treatment scaffold to the securement member.

[0017] Alternatively or additionally, the method may further include selecting an appropriately sized treatment scaffold out of a plurality of differently sized treatment scaffolds.

[0018] Alternatively or additionally, the securement member may include an elongate member, a cross member extending laterally from the elongate member, and a plurality of fork members extending from the cross member, the plurality of fork members extending in a plane transverse to a plane extending through the elongate member and the cross member.

[0019] Alternatively or additionally, the step of deploying the treatment scaffold within the meniscal tear may include substantially aligning collagen fibers of the treatment scaffold with collagen fibers of the meniscus.

[0020] Alternatively or additionally, the one or more sutures may extend through the treatment implant.

[0021] Another example may be found in a device adapted for securing a meniscal implant for delivery to a meniscal tear. The device includes an elongate member, a cross member including a first cross member segment extending laterally from the elongate member in a first direction and a second cross member segment extending laterally from the elongate member in a second direction, and a plurality of fork members extending from the cross member, the plurality of fork members extending in a plane transverse to a plane extending through the elongate member and the cross member, at least some of the plurality of fork members adapted to extend into a meniscal implant. The first cross member segment and the second cross member segment are each adapted to fold

over on themselves.

[0022] Alternatively or additionally, the cross member and the plurality of fork members may each include Nitinol.

[0023] Alternatively or additionally, the device may be constructed by processing a flat strip of Nitinol to form a blank including the elongate member, the cross member and the plurality of fork members, the blank extending within a single plane. The blank is shape set, with the plurality of fork members extending in the plane transverse to the plane extending through the elongate member and the cross member. Shape setting the blank further includes providing the delivery device with a remembered shape in which the first cross member segment extends laterally from the elongate member in a first direction, the second cross member segment extends laterally from the elongate member in a second direction, and the plurality of fork members extends in a plane transverse to a plane extending through the elongate member and the cross member.

[0024] The above summary of some embodiments is not intended to describe each disclosed embodiment or every implementation of the present disclosure. The Figures, and Detailed Description, which follow, more particularly exemplify these embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The disclosure may be more completely understood in consideration of the following detailed description in connection with the accompanying drawings, in which:

[0026] FIGS. 1A and 1B are perspective views of illustrative collagen scaffolds;

[0027] FIG. 2 is a side view of an illustrative blank that may be used to form an illustrative securement member;

[0028] FIG. 3 is a perspective view of an illustrative securement member formed from the blank of FIG. 2;

[0029] FIGS. 4 through 6 are perspective views of a process of forming an illustrative scaffold assembly; and

[0030] FIGS. 7 through 14 are perspective views of a process of implanting a treatment scaffold within a meniscal tear.

[0031] While the disclosure is amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit the disclosure to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosure.

DETAILED DESCRIPTION

[0032] For the following defined terms, these definitions shall be applied, unless a different definition is given in the claims or elsewhere in this specification.

[0033] All numeric values are herein assumed to be modified by the term “about”, whether or not explicitly indicated. The term “about” generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (e.g., having the same function or result). In many instances, the terms “about” may include numbers that are rounded to the nearest significant figure.

[0034] The recitation of numerical ranges by endpoints includes all numbers within that range (e.g. 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5).

[0035] As used in this specification and the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.

[0036] It is noted that references in the specification to “an embodiment”, “some embodiments”, “other embodiments”, etc., indicate that the embodiment described may include one or more particular features, structures, and/or characteristics. However, such recitations do not necessarily mean that all embodiments include the particular features, structures, and/or characteristics. Additionally, when particular features, structures, and/or characteristics are described in connection with one embodiment, it should be understood that such features, structures, and/or characteristics may also be used connection with other embodiments whether or not explicitly described unless clearly stated to the contrary.

[0037] The following detailed description should be read with reference to the drawings in which similar elements in different drawings are numbered the same. The drawings, which are not necessarily to scale, depict illustrative embodiments and are not intended to limit the scope of the disclosure.

[0038] Meniscal implants such as treatment scaffolds may be provided in a variety of sizes. FIG. 1A shows an example of an illustrative treatment scaffold **10** having a length L, a width W and a thickness T. In some instances, L may vary from 10 to 40 millimeters in length, W may vary from 3 to 10 millimeters in width and T may vary from 1 to 5 millimeters thick. In some cases, the treatment scaffold **10** may be manufactured having particular dimensions, and prior to use, the physician or other professional may trim the treatment scaffold **10** in one or more dimensions in order to make the treatment scaffold **10** be an appropriate size for repairing a particular meniscal tear for a particular patient.

[0039] In some instances, the treatment scaffold **10** may be formed of a variety of different types of fibers. In some cases, the treatment scaffold **10** may include collagen fibers, particularly since the meniscus is largely formed of collagen and water. Some estimates place the meniscus as being about 72% water and about 22 percent collagen, with minor amounts of other materials. The treatment scaffold **10** may include collagen fibers that are loosely aligned along a long axis of the treatment scaffold **10**. This may be beneficial, as the collagen fibers forming the C-shaped meniscus are themselves loosely aligned around the “C” shape. As a result of aligning the treatment scaffold **10** with its long dimension extending along the arcuate direction of the “C” shape, the collagen fibers within the treatment scaffold **10** may be oriented generally parallel to the collagen fibers forming the meniscus such that the collagen fibers within the treatment scaffold **10** are generally aligned with the native collagen fibers forming the meniscus. In some instances, the collagen fibers within the treatment scaffold **10** may be oriented generally parallel to the collagen fibers forming the meniscus, such that the collagen fibers within the treatment scaffold **10** are substantially aligned with the native collagen fibers forming the meniscus. In some cases, this can help with healing. In some cases, the treatment scaffold **10** may include loosely oriented type 1 collagen fibers with high porosity. It will be appreciated that the treatment scaffold **10**, once disposed within the meniscal tear, may promote healing by providing a scaffold for biological material to fill up.

[0040] While not shown, the treatment scaffold **10** may include one or more biological factors, such as but not limited to substances that promote healing even in the absence of blood supply. These biological factors may be included particularly when the meniscal tear being treated is in the white zone of the meniscus, or sometimes even when the meniscal tear being treated is in the red-white zone of the meniscus. In some cases, repairs to the white zone of the meniscus do not heal well because of a lack of blood flow within the white zone of the meniscus. In some cases, repairs to the red-white zone of the meniscus also suffer from limited blood flow, even though the red-white zone can have at least some blood flow, especially relative to the white zone. It is thought that inclusion of a scaffold such as a collagen scaffold in treating meniscal tears may improve healing, even in the absence of adequate blood flow. Inclusion of biological factors may aid in healing.

[0041] In some cases, and as shown in FIG. 1B, the physician or other professional may be

provided with a kit **12** that includes several different treatment scaffolds that may include or are made of collagen, including a collagen scaffold **10a**, a collagen scaffold **10b** and a collagen scaffold **10c**. While described with respect to collagen, other materials are also contemplated. While the kit **12** is shown as including a total of three collagen scaffolds **10a**, **10b** and **10c**, this is merely illustrative, as the kit **12** may include any number of collagen scaffolds. In some cases, the kit **12** may include several different collagen scaffolds of differing sizes, and may also include one or more larger collagen scaffolds that are intended to be cut down prior to use. In FIG. **1B**, the collagen scaffold **10a** is shown as being smaller than the collagen scaffold **10** shown in FIG. **1A**. The collagen scaffold **10b** is shown as having a larger width than the collagen scaffold **10a**, but a shorter length. The collagen scaffold **10c** is shown as having a length that is greater than the length **L** of the collagen scaffold **10**, and perhaps a slightly larger width as well.

[0042] While drawn as rectilinear, it will be appreciated that this is merely illustrative, as the collagen scaffold **10**, **10a**, **10b** and **10c** may take any desired shape to best fit at a desired treatment site such as but not limited to a meniscal tear repair. In some cases, the collagen scaffold **10**, **10a**, **10b** and **10c** may be largely rectilinear, with rounded over corners, for example. In some cases, the collagen scaffold **10**, **10a**, **10b** and **10c** may be at least partially ovoid. The first primary surface **104** and the second primary surface **106** may be planar or curved. In some cases, the collagen scaffold **10**, **10a**, **10b** and **10c** may curve around a treatment site. These are just examples.

[0043] The collagen used to form the collagen scaffold **10**, **10a**, **10b** and **10c** may come from any of a variety of different sources. Collagen is a main structural protein in the extracellular matrix found in various connective tissues and thus can be obtained from various animals. For example, the collagen used to form the collagen scaffold **10**, **10a**, **10b** and **10c** may be bovine-based, i.e., from cows. In some cases, the collagen used to form the collagen scaffold **10**, **10a**, **10b** and **10c** may come from bovine tendon material. As an illustrative but non-limiting example, bovine Achilles tendon is digested down to highly purified collagen, and is reconstituted into a sheet by spinning the fibers around a mandrel. The collagen scaffold **10**, **10a**, **10b** and **10c** may be formed of collagen that has been obtained in other methods as well.

[0044] In some instances, the collagen scaffold **10**, **10a**, **10b** and **10c** may include a combination of collagen fibers or material and other biocompatible fibers or materials. For example, the collagen scaffold **10**, **10a**, **10b** and **10c** may include a combination of collagen fibers and bioabsorbable polymer fibers, if desired.

[0045] While the collagen scaffold **10**, **10a**, **10b** and **10c** may be formed having a variety of different porosity levels (defined as relative amount of void space to solid material), in some cases the collagen scaffold **10**, **10a**, **10b** and **10c** may have a porosity of at least 50 percent or more, at least 60 percent or more, at least 70 percent or more, or at least 80 percent or more. In some cases, the scaffold **102** may have a porosity of 60 percent to 90 percent, 70 percent to 90 percent, 80 percent to 90 percent, or 85 percent to 90 percent.

[0046] Another way to define the collagen scaffold **10**, **10a**, **10b** and **10c** is in terms of average pore size. Pore size refers to a diameter of voids, or empty spaces, formed within the collagen scaffold **10**, **10a**, **10b** and **10c**. Average pore size, accordingly, refers to an average diameter of these voids. In some cases, the collagen scaffold **10**, **10a**, **10b** and **10c** may have an average pore size of 20 microns or greater, or 30 microns or greater, or 40 microns or greater, or 50 microns or greater, or 60 microns or greater, or 70 microns or greater, or 80 microns or greater, or 90 microns or greater. In some cases, the collagen scaffold **10**, **10a**, **10b** and **10c** may have an average pore size that is in a range of 100 microns to 500 microns, 100 microns to 400 microns, 100 microns to 300 microns, 100 microns to 150 microns, or 200 microns to 400 microns, for example.

[0047] Once the collagen scaffold **10**, **10a**, **10b** and **10c** has been appropriately trimmed for use, or has been appropriately selected to have the appropriate size, the collagen scaffold **10**, **10a**, **10b** and **10c** can be delivered. The collagen scaffold **10**, **10a**, **10b** and **10c** may be delivered in a variety of ways. For example, the collagen scaffold **10**, **10a**, **10b** and **10c** may simply be delivered to a

meniscal tear using a grasping device and optionally a standard cannula. In some cases, such as for radial tears and/or complex tear geometries, delivering the collagen scaffold **10**, **10a**, **10b** and **10c** with a graspers may be optimal. In some instances, the collagen scaffold **10**, **10a**, **10b** and **10c** may be delivered using a delivery device that includes a sheath that extends over the scaffold **10**, **10a**, **10b** and **10c** in order to hold the scaffold **10**, **10a**, **10b** and **10c** in a collapsed configuration for delivery. Once the scaffold **10**, **10a**, **10b** and **10c** has been delivered to a position proximate a meniscal tear, the scaffold **10**, **10a**, **10b** and **10c** may be deployed from the delivery device by retracting the sheath extending over the scaffold **10**, **10a**, **10b** and **10c**, or perhaps by pushing the scaffold **10**, **10a**, **10b** and **10c** distally out of the sheath. Other systems for delivering the scaffold **10**, **10a**, **10b** and **10c** are contemplated. In some cases, however, the collagen scaffold **10**, **10a**, **10b** and **10c** may be delivered using a delivery device such as that described with respect to FIGS. 2 through 14.

[0048] FIG. 2 shows an illustrative blank **14** that may be processed from a flat sheet or strip of Nitinol. In some cases, the blank **14** may be laser cut, stamped or machined, for example. Other cutting processes are contemplated. As can be seen, the blank **14** has been cut to include an elongate member **16**, a cross member **18** that extends laterally from the elongate member **16**, including a first cross member segment **18a** that extends laterally from the elongate member **16** in a first direction indicated by an arrow **20a**, and a second cross member segment **18b** that extends laterally from the elongate member **16** in a second direction indicated by an arrow **20b**. In some cases, as shown, the second direction is opposite the first direction. The blank **14** includes a number of forks **22**, individually labeled as **22a**, **22b** and **22c**. While a total of three forks **22** are shown, this is merely illustrative, as the blank **14** may include any number of forks **22**, such as a total of two forks **22**, or four or more forks **22**. As formed, the blank **14** may be considered as being planar, with the elongate member **16**, the cross member **18** and the forks **22** being disposed within a single plane that is parallel to the page.

[0049] FIG. 3 shows a securement member **24** of an implant delivery device that may be formed by shape-setting the blank **14**. As shown in FIG. 3, the securement member **24** has been processed such that the forks **22** extend downward (in the illustrated orientation) from the cross member **18**. In some cases, the forks **22** may be considered as extending within a plane that is orthogonal, or at least substantially orthogonal, to a plane that may be imagined as extending through the elongate member **16** and the cross member **18**. In some cases, each of the forks **22** may be considered as being parallel with each of the other forks **22**. In some cases, one or more of the forks **22** may lie either in front of or behind the imaginary plane extending through the forks **22** (and orthogonal or at least substantially orthogonal to the imaginary plane extending through the elongate member **16** and the cross member **18**). The securement member **24** may be considered as having a remembered shape that matches that shown in FIG. 3. Even if the securement member **24** is moved out of the shape shown in FIG. 3, the securement member **24** will attempt to regain the shape shown in FIG. 3 anytime that the securement member **24** is not constrained from doing so.

[0050] The securement member **24** may have dimensions that are determined at least in part upon the final dimensions of the collagen scaffold **10**, **10a**, **10b** and **10c**. For example, the forks **22** may have a length that is at least as long as the width **W** of the collagen scaffold **10**, **10a**, **10b** and **10c**. The cross member **18** may have a length that is about as long as the length **L** of the collagen scaffold **10**, **10a**, **10b** and **10c**. If the collagen scaffold **10**, **10a**, **10b** and **10c** is particularly thin, this may mean that the forks **22** have a relatively smaller diameter, for example. In some cases, different size securement members **24** may be selected, based on the overall dimensions of the collagen scaffold **10**, **10a**, **10b** and **10c**.

[0051] FIGS. 4 through 6 show an example of securing the collagen scaffold **10** (which could be any one of the collagen scaffolds **10**, **10a**, **10b** and **10c** described herein, or variations thereof) relative to the securement member **24** of an implant delivery device. For example, the forks **22** of the securement member **24** may be secured to the collagen scaffold **10**. As shown in FIG. 4, in

some instances the securement member **24** may be secured to the collagen scaffold **10** by penetrating the forks **22** into the collagen scaffold **10**, such as by threading the forks **22** through the collagen scaffold **10**. As shown, each of the forks **22** appear to have been threaded into the collagen scaffold **10**, with the material forming the collagen scaffold **10** alternatively weaving in front of and behind each of the forks **22**. This is merely illustrative, as in some cases, and depending on the particular dimensions of the collagen scaffold **10**, each of the forks **22** may simply be inserted into spaces within the collagen scaffold **10**, without threading the forks **22** into the collagen scaffold **10**, or otherwise secured to the collagen scaffold **10**.

[0052] Once secured to the securement member **24** of the delivery device, the collagen scaffold **10**, as well as the securement member **24** may be collapsed, or otherwise manipulated into a collapsed, delivery configuration. For example, in FIG. 5, the collagen scaffold **10** has been folded over on itself. This involves the cross members **18a** and **18b** each being adapted to fold over on themselves. In FIG. 6, and after being folded, the collagen scaffold **10** has been rolled up. This involves bending each of the forks **22**. First folding, and then rolling up the collagen scaffold **10** allows the collagen scaffold **10** and the securement member **24** to, in combination, take up a reduced volume, which can help with disposing the collagen scaffold **10** and the securement member **24** in a delivery device, as will be discussed. It will be appreciated that bending the cross member **18** (in folding the collagen scaffold **10**) and bending the forks **22** (in rolling up the folded collagen scaffold **10**) cause the securement member **24** to be deformed from its remembered shape, or configuration. The folded and/or rolled up collagen scaffold **10**, in combination with the similarly deformed securement member **24**, may be considered as forming a scaffold assembly **26**. In some cases, the scaffold assembly **26** may be held in this deformed, collapsed shape by virtue of a sheath or other structure temporarily placed over the scaffold assembly **26**. In some cases, the scaffold assembly **26** may be assembled prior to a surgical procedure, such as subsequent to selecting an appropriately sized collagen scaffold **10**, or after trimming a collagen scaffold to be an appropriate size. In some cases, the scaffold assembly **26** may be provided already assembled, and the physician or other professional merely needs to load the scaffold assembly **26** into a delivery device. In some cases, the physician or other professional may select a particular scaffold assembly **26** out of several different scaffold assemblies, where each of the several different scaffold assemblies include a particular size of collagen scaffold. This allows use of an appropriately sized collagen scaffold for the particular meniscal tear to be repaired. A first scaffold assembly **26** may include a small-sized collagen scaffold, a second scaffold assembly **26** may include a medium-sized collagen scaffold and a third scaffold assembly **26** may include a large-sized collagen scaffold, for example.

[0053] FIG. 7 schematically shows a delivery device **30** into which the scaffold assembly **26** has been loaded. The delivery device **30** includes a handle **32** including an actuation mechanism, such as a trigger **34** that is adapted to be movable back and forth within the handle **32**. A sheath **36** is disposed over the scaffold assembly **26** in order to hold the scaffold assembly **26** in its current, collapsed configuration. In some cases, the scaffold assembly **26** would be provided already loaded into the delivery device **30**. If the scaffold assembly **26** is to be cut to size before use, the physician would weave or spear the fork tines through the scaffold assembly **26**, then roll the fork-scaffold assembly up, followed by advancing the trigger **34** to advance the sheath **36** over the rolled-up scaffold. In some cases, the sheath **36** may be clear or translucent, such that components within the sheath **36** may be visualized through the sidewall of the sheath **36**. The delivery device **30** may also include an elongate shaft **38**. The scaffold assembly **26**, including the securement member **24** and the collagen scaffold **10** may be secured to the distal end of the elongate shaft **38**. The elongate shaft **38** may be formed of any desired material, such as a malleable metallic material, for instance stainless steel or martensitic nitinol, for example.

[0054] In some cases, the elongate shaft **38** may initially be straight, and may be curved or otherwise bent in order to best reach a location of a particular meniscal tear within a particular

patient. In some cases, the elongate shaft **38** may be adapted to be able to be manually bent by hand, which allows the physician or other professional to manipulate the elongate shaft **38** within their hands in order to bend the elongate shaft **38** to a desired curved shape during a surgical procedure. In some cases, the physician or other professional may utilize a mold in bending the elongate shaft **38** to a desired curved shape. In some cases, utilizing a mold may provide a smoother bend to the elongate shaft **38**, with less chance of kinking the elongate shaft **38**. US Pat. App. Pub. No. 2022/0161306, the disclosure of which is incorporated herein by reference, provides an example of a suitable mold.

[0055] A cannula **40** may extend over the elongate shaft **38**. The cannula **40** may have a proximal end located proximate to the handle **32** and extend distally therefrom to a distal end of the cannula **40**. The cannula **40** may extend distal of the elongate shaft **38** and cover the collapsed scaffold assembly **26** for delivery to an anatomical location.

[0056] In FIG. 7, the delivery device **30** is shown next to a schematically illustrated meniscus **42** that has a meniscus tear **44**. It will be appreciated that the meniscus tear **44**, as shown, is a vertical longitudinal tear. Other types of meniscus tears are known and contemplated herein.

[0057] In FIG. 8, the cannula **40** can be seen as being partially removed. In some cases, the cannula **40** is a tear-away cannula that can simply be peeled away and disposed of. For example, a proximal end portion of the cannula **40** may be grasped by a user and pulled relative to the handle **32** and the elongate shaft **38**, thereby removing the cannula from the delivery device **30**. The cannula **40** may be removed once the delivery device **30** has entered the joint space near the meniscus **42**.

Thereafter, the delivery device **30** may be advanced in the joint space to navigate the scaffold assembly **26** to the location of the meniscus tear **44**. This is shown in FIG. 9, where the distal end of the delivery device **30**, including the scaffold assembly **26** and the sheath **36** surrounding the scaffold assembly **26**, are positioned proximate the meniscus **42** and the meniscus tear **44**.

[0058] Once the scaffold assembly **26** has been positioned proximate the meniscus tear **44**, the actuation mechanism provided with the handle **32** may be actuated to deploy the securement member **24** and the collagen scaffold **10** from the distal opening of the sheath **36**. For example, as shown in FIG. 10, the trigger **34** has been moved proximally relative to the handle **32**, thereby withdrawing the sheath **36** that was previously overlaying the scaffold assembly **26**. As a result, the scaffold assembly **26**, including the collagen scaffold **10** and the securement member **24**, may be deployed out of the sheath **36** such that the securement member **24** may revert back to its remembered configuration in order to open or unfold the collagen scaffold **10** into a flattened configuration. The collagen scaffold **10** may then be inserted into the meniscus tear **44** in an expanded, flattened configuration. In some instances, the collagen fibers within the collagen scaffold **10** may be oriented generally parallel to the collagen fibers forming the meniscus such that the collagen fibers within the collagen scaffold **10** are generally aligned with the native collagen fibers forming the meniscus. In some instances, the collagen fibers within the collagen scaffold **10** may be oriented generally parallel to the collagen fibers forming the meniscus, such that the collagen fibers within the collagen scaffold **10** are substantially aligned with the native collagen fibers forming the meniscus.

[0059] In some cases, the securement member **24** may be used to hold the collagen scaffold **10** within the meniscus tear **44** as the meniscus tear **44** is repaired, such as by suturing the meniscus tear **44** closed with one or more sutures while retaining the collagen scaffold **10** within the meniscus tear **44**. In other instances, the securement member may be withdrawn from the collagen scaffold **10** in order to get the securement member **24** out of the way prior to suturing. In some cases, as shown in FIG. 12, a secondary device **46** may be used to assist with the repair of the meniscal tear **44**. For example a forceps or other grasping device may be used to grasp and/or manipulate the collagen scaffold **10** to correctly position and/or retain the collagen scaffold **10** within the meniscus tear **44**. In some instances one or more of the sutures may extend through the collagen scaffold **10** as the sutures are sutured through meniscal tissue on both sides of the

meniscus tear **44**. In other instances, the sutures may not extend through the collagen scaffold **10** (e.g., the sutures may extend across the meniscus tear **44** above the collagen scaffold **10**) as the sutures are sutured through the meniscal tissue on both sides of the meniscus tear **44**.

[0060] In some cases, the securement member **24** may be withdrawn back into the sheath **36** by actuating the actuation mechanism, such as moving the trigger **34** distally as shown in FIG. **14**. This causes the sheath **36** to advance distally over the securement member **24**, which protects the securement member **24** from contacting or damaging any anatomy during removal of the delivery device **30** from the joint space. FIG. **13** shows the repaired meniscal tear **44** with several sutures **48** that have been placed in order to secure the position of the collagen scaffold **10** within the meniscal tear **44**. In some cases, the sutures **48** may be placed after the securement member **24** has been withdrawn from the collagen scaffold **10**. In some cases, the sutures **48** may be placed while the securement member **24** remains within the collagen scaffold **10**. The sutures **48** may be placed using any desired technique, process or equipment. In some instances, the sutures **48** may be passed through the thickness of the collagen scaffold **10** as the sutures are passed through the meniscal tissue to close the meniscal tear **44**. While sutures are shown, other tissue fasteners are contemplated. In some cases, the sutures **48** may be placed using the FAST-FIT FLEX meniscus repair device available commercially from Smith & Nephew. In some cases, the secondary device **46** may represent the FAST-FIT FLEX meniscus repair device, for example. Additional suturing devices are disclosed in U.S. Pat. Nos. 7,887,551 and 8,986,343, the disclosures of which are herein incorporated by reference.

[0061] It should be understood that this disclosure is, in many respects, only illustrative. Changes may be made in details, particularly in matters of shape, size, and arrangement of steps without exceeding the scope of the disclosure. This may include, to the extent that it is appropriate, the use of any of the features of one example embodiment being used in other embodiments. The disclosure's scope is, of course, defined in the language in which the appended claims are expressed.

Claims

1. A method for repairing a meniscal tear within a patient's meniscus, the method comprising: securing a meniscal implant relative to a delivery device to form a scaffold delivery assembly; delivering the scaffold delivery assembly to a position proximate the meniscal tear; inserting the meniscal implant into the meniscal tear; and holding the meniscal implant within the meniscal tear by placing one or more fasteners extending across the meniscal tear.
2. The method of claim 1, further comprising separating the meniscal implant from the delivery device after inserting the meniscal implant into the meniscal tear.
3. The method of claim 2, wherein the method further comprises one of: intra-operatively cutting the meniscal implant to an appropriate size prior to securing the meniscal implant to the delivery device; or selecting an appropriately sized meniscal implant out of a plurality of differently sized meniscal implants.
4. The method of claim 1, wherein the delivery device comprises: an elongate member; a cross member including a first cross member segment extending laterally from the elongate member in a first direction and a second cross member segment extending laterally from the elongate member in a second direction; and a plurality of fork members extending from the cross member, the plurality of fork members extending in a plane transverse to a plane extending through the elongate member and the cross member.
5. The method of claim 1, further comprising: intra-operatively imparting a bend to an elongate shaft of the delivery device prior to delivering the scaffold delivery assembly to the position proximate the meniscal tear.
6. The method of claim 1, wherein the step of inserting the meniscal implant into the meniscal tear

includes substantially aligning collagen fibers of the meniscal implant with collagen fibers of the meniscus.

7. A method for repairing a meniscal tear within a patient's meniscus, the method comprising: inserting a meniscal implant secured to a distal end region of a delivery device into the meniscal tear; and placing one or more fasteners extending across the meniscal tear through the meniscal implant to hold the meniscal implant within the meniscal tear; thereafter, decoupling the delivery device from the meniscal implant.

8. The method of claim 7, wherein the delivery device comprises: an elongate member; a cross member including a first cross member segment extending laterally from the elongate member in a first direction and a second cross member segment extending laterally from the elongate member in a second direction; and a plurality of fork members extending from the cross member, the plurality of fork members extending in a plane transverse to a plane extending through the elongate member and the cross member.

9. The method of claim 8, further comprising: intra-operatively attaching the meniscal implant to the plurality of fork members prior to inserting the meniscal implant into the meniscal tear.

10. The method of claim 8, further comprising: intra-operatively imparting a bend to the elongate shaft of the delivery device prior to inserting the meniscal implant into the meniscal tear.

11. The method of claim 7, wherein the step of inserting the meniscal implant into the meniscal tear includes substantially aligning collagen fibers of the meniscal implant with collagen fibers of the meniscus.

12. The method of claim 7, wherein the method further comprises one of: intra-operatively cutting the meniscal implant to an appropriate size prior to inserting the meniscal implant into the meniscal tear; or intra-operatively selecting an appropriately sized meniscal implant out of a plurality of differently sized meniscal implants prior to inserting the meniscal implant into the meniscal tear.

13. A method for repairing a meniscal tear within a patient's meniscus, the method comprising: inserting a meniscal implant formed of collagen fibers into the meniscal tear such that the collagen fibers of the meniscal implant are substantially aligned with collagen fibers of the meniscus; and placing one or more fasteners extending across the meniscal tear through the meniscal implant to hold the meniscal implant within the meniscal tear.

14. The method of claim 13, wherein the meniscal implant is inserted into the meniscal tear using a delivery device having the meniscal implant coupled to a distal end thereof.

15. The method of claim 14, wherein the delivery device comprises: an elongate member; a cross member including a first cross member segment extending laterally from the elongate member in a first direction and a second cross member segment extending laterally from the elongate member in a second direction; and a plurality of fork members extending from the cross member, the plurality of fork members extending in a plane transverse to a plane extending through the elongate member and the cross member.

16. The method of claim 15, wherein the plurality of fork members include at least three forks.

17. The method of claim 15, further comprising: intra-operatively attaching the meniscal implant to the plurality of fork members prior to inserting the meniscal implant into the meniscal tear.

18. The method of claim 14, further comprising: intra-operatively imparting a bend to an elongate shaft of the delivery device prior to inserting the meniscal implant into the meniscal tear.

19. The method of claim 14, further comprising: advancing the meniscal implant to the meniscal tear with the meniscal implant in a collapsed configuration within a sheath; and unfolding the meniscal implant into a flattened configuration at the meniscal tear by deploying the meniscal implant from a distal end of the sheath.

20. The method of claim 13, wherein the method further comprises one of: intra-operatively cutting the meniscal implant to an appropriate size prior to inserting the meniscal implant into the meniscal tear; or intra-operatively selecting an appropriately sized meniscal implant out of a plurality of differently sized meniscal implants prior to inserting the meniscal implant into the meniscal tear.

